VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 4

Query Languages

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Content

- Introduction
- Define relation schemas
- Manipulate on relation schemas
- Summary

Introduction

- Study database programming
  - How the user can ask queries of the database
    - Select
  - How the user can modify contents of the database
    - Insert, delete and update

- Relational model
  - (1) Relational Algebra
    - Present a query by expressions
  - (2) Relational Calculus
    - Present the result of a query
  - (3) SQL (Structured Query Language)

(1) Relational algebra language

- **How** to execute query *operations* (in what order)
  - Selection
  - Projection
  - Cartesian production
  - Join operation
  - Set operations
  - Division operation
  - Modification operations

- Difficult for users

(2) Relational calculus language

- The formal query language
- Introduced by Codd in 1972, "Data Base Systems", Prentice Hall, p33-98

- Nonprocedural language
  - Calculus expression specifies *what is to be retrieved* rather than *how to retrieve*
- One declarative expression to specify a retrieval request
  - There is no description of how to evaluate query
- A calculus expression may be written in different way
  - The way it is written has no bearing on how a query should be evaluated

(3) SQL

- Structured Query Language
  - High level declarative language interface
  - The user only specifies what the result is to be

  - Developed at IBM Research (1970s)
  - Also pronounced SEQUEL
  - Expanded to a standard by ANSI
    - SQL1: SQL-86
    - SQL2: SQL-92
    - SQL3: SQL-99, SQL-2003, SQL-2006, SQL-2008

(3) SQL

- Based on

Relational algebra operations	+	Some supplements

- Allow a table to have two or more tuples that are identical in all their attribute values
- Not a set of tuples, but a multiset or bag

(3) SQL

- Includes
  - Data definition language (DDL)
  - Data manipulation language (MDL)
  - View definition language (VDL)
  - Integrity constraint

  - Security and authorization
  - Transaction control

  - Rules for embedding SQL into programming languages

Content

- Introduction
- Define relation schemas in SQL
  - Data definition
  - Data type
- Manipulate on relation schemas

Data definition

- Describes the structure of information in the DB
  - Schema for the relation
  - Domain of each attribute
  - Integrity constraint
  - Index on each relation
- In SQL
  - CREATE TABLE
  - DROP TABLE
  - ALTER TABLE
  - CREATE DOMAIN
  - CREATE DATABASE…

Data type

- Numeric
  - INTEGER
  - SMALLINT
  - NUMERIC, NUMERIC(p), NUMERIC(p,s)
  - DECIMAL, DECIMAL(p), DECIMAL(p,s)
  - REAL
  - DOUBLE PRECISION
  - FLOAT, FLOAT(p)

Data type

- Character string
  - CHARACTER, CHARACTER(n)
  - CHARACTER VARYING(x)

- Bit string
  - BIT, BIT(x)
  - BIT VARYING(x)

- Datetime
  - DATE
  - TIME
  - TIMESTAMP

Create table command

- Define a new relation by giving
  - A name
  - Attributes
    - Names
    - Data types
    - Integrity constraints on attributes
- Syntax

```
CREATE TABLE <Table_name> (
        <Column_name> <Data_type> [<Constraint>],
        <Column_name> <Data_type> [<Constraint>],
        …
        [<Constraint>]
)
```

Example

```
CREATE TABLE EMPLOYEE (
        LName VARCHAR(10),
        Minit CHAR(1),
        FName VARCHAR(10),
        SSN CHAR(9),
        BDate DATETIME,
        Address VARCHAR(50),
        Sex CHAR(6),
        Salary INT,
        SuperSSN CHAR(9),
        DNo INT
)
```

Create table command

- Basic constraints
  - NOT NULL
  - NULL
  - UNIQUE
  - DEFAULT
  - PRIMARY KEY
  - FOREIGN KEY / REFERENCES
  - CHECK

- Give a name to constraints

CONSTRAINT <Constraint_name> <Constraint>

Example – Constraint

```
CREATE TABLE EMPLOYEE (
        LName VARCHAR(10) NOT NULL,
        Minit VARCHAR(20) NOT NULL,
        FName VARCHAR(10) NOT NULL,
        SSN CHAR(9) PRIMARY KEY,
        BDate DATETIME,
        Address VARCHAR(50),
        Sex CHAR(6) CHECK (Sex IN ( 'Male' ,  'Female' )),
        Salary INT DEFAULT (10000),
        SuperSSN CHAR(9),
        DNo INT
)
```

Example – Constraint

```
CREATE TABLE DEPARTMENT (
        DName VARCHAR(20) UNIQUE,
        DNumber INT NOT NULL,
        MgrSSN CHAR(9),
        MgrStartDate DATETIME DEFAULT (GETDATE())
)

CREATE TABLE WORKS_ON (
        ESSN CHAR(9) FOREIGN KEY (ESSN)
                REFERENCES EMPLOYEE(SSN),
        PNo INT REFERENCES PROJECT(PNumber),
        Hours DECIMAL(3,1)
)
```

Example – Constraint name

```
CREATE TABLE EMPLOYEE (
        LName VARCHAR(10) CONSTRAINT EM_LName_NN NOT NULL,
        MName VARCHAR(20) NOT NULL,
        FName VARCHAR(10) NOT NULL,
        SSN CHAR(9) CONSTRAINT EM_SSN_PK PRIMARY KEY,
        BDate DATETIME,
        Address VARCHAR(50),
        Sex CHAR(6) CONSTRAINT EM_Sex_CHK
                CHECK (Sex IN ( 'Male' ,  'Female' )),
        Salary INT CONSTRAINT EM_Salary_DF DEFAULT (10000),
        SuperSSN CHAR(9),
        DNo INT
)
```

Example – Constraint name

```
CREATE TABLE WORKS_ON (
        ESSN CHAR(9),
        PNo INT,
        Hours DECIMAL(3,1),
        CONSTRAINT WO_SSN_PNo_PK PRIMARY KEY (ESSN, PNo),
        CONSTRAINT WO_SSN_FK FOREIGN KEY (ESSN)
                REFERENCES EMPLOYEE(SSN),
        CONSTRAINT WO_PNo_FK FOREIGN KEY (PNo)
                REFERENCES PROJECT(PNumber)
)
```

Alter table command

- Is used for modification
  - The structure of tables
  - Integrity constraints

- Columns

```
ALTER TABLE <Table_name> ADD COLUMN
          <Column_name> <Data_type> [<Constraint>]
```

```
ALTER TABLE <Table_name> DROP COLUMN <Column_name>
```

```
ALTER TABLE <Table_name> ALTER COLUMN
          <Column_name> <New_data_type>
```

Alter table command

- Constraints

```
ALTER TABLE <Table_name> ADD
          CONSTRAINT <Constraint_name> <Constraint>,
          CONSTRAINT <Constraint_name> <Constraint>,
          …
```

```
ALTER TABLE <Table_name> DROP <Constraint_name>
```

Example

```
ALTER TABLE EMPLOYEE ADD
        JobTitle CHAR(20)

ALTER TABLE EMPLOYEE ALTER COLUMN
        JobTitle CHAR(50)

ALTER TABLE EMPLOYEE DROP COLUMN JobTitle
```

Example

```
CREATE TABLE DEPARTMENT (
        DName VARCHAR(20),
        DNumber INT NOT NULL,
        MgrSSN CHAR(9),
        MgrStartDate DATETIME
)
ALTER TABLE DEPARTMENT ADD
        CONSTRAINT DE_DNumber_PK PRIMARY KEY (DNumber),
        CONSTRAINT DE_MgrSSN_FK FOREIGN KEY (MgrSSN)
                REFERENCES EMPLOYEE(SSN),
        CONSTRAINT DE_MgrStartDate_DF DEFAULT (GETDATE())
                FOR (MgrStartDate),
        CONSTRAINT DE_DName_UN UNIQUE (DName)
```

Drop table command

- Is used for deleting the structure of tables
  - All the data in a table are also deleted

- Syntax

  DROP TABLE <Table_name>

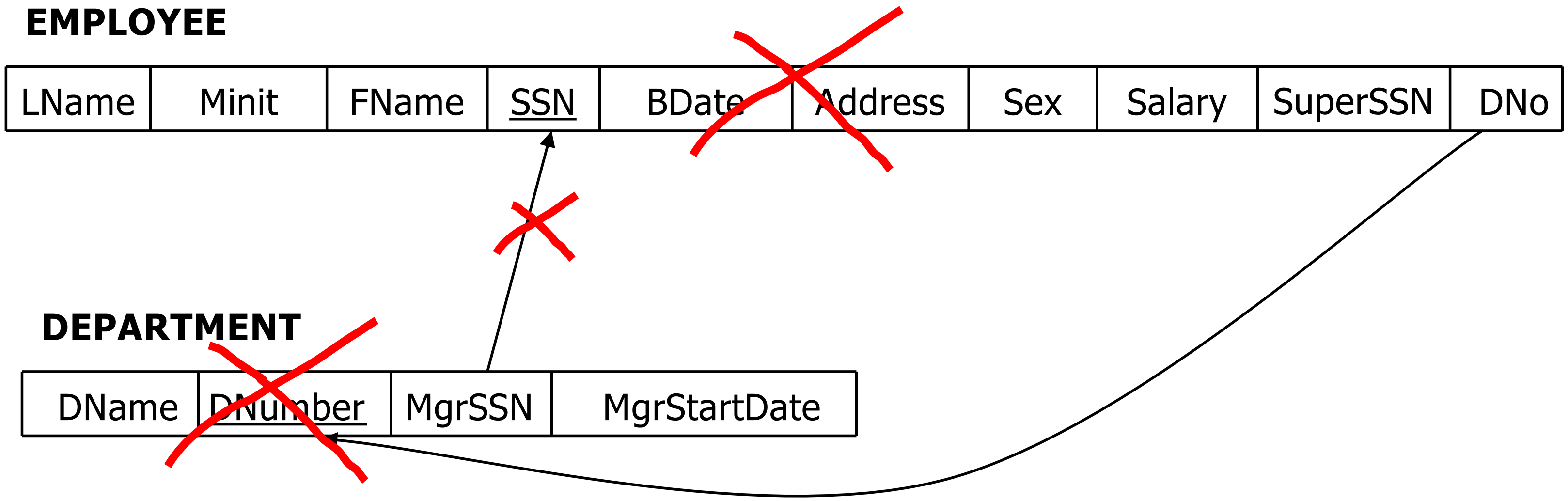
- Example

  DROP TABLE EMPLOYEE

  DROP TABLE DEPARTMENT

  DROP TABLE WORKS_ON

Example


EMPLOYEE
LName
Minit
FName
SSN
BDate
Address
Sex
Salary
SuperSSN
DNo
DEPARTMENT
DName
DNumber
MgrSSN
MgrStartDate

Content

- Introduction
- Define relation schemas
- Manipulate on relation schemas
  - **Data modification**
  - Data retrieval
    - Basic query
      - One schema
      - Two or many schemas
    - Advanced query
- Summary

Introduction

- Consider manipulations on the relation EMPLOYEE
  - Add a new employee
  - Move the employee “John Smith” to the department 1
  - List names and birth dates of employees whose salary are over 20,000

FName	LName	BDate	Address	Sex	Salary	DNo
John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
Joyce	English	07/31/1972	5631 Rice, Houston, TX	Female	25000	5
James	Borg	11/10/1937	450 Stone, Houston, TX	Male	55000	1

Introduction

- Data manipulation language is used for
  - Modifying data in a database
    - Eg. Add a new employee
    - Eg. Move the employee "John Smith" to the department 1

  → Relational algebra and SQL

  - Retrieving information from a database
    - Eg. List names and birth dates of employees whose salary are over $20,000

  → Relational algebra, relational calculus and SQL

Data modification in SQL

- Insert command
  - Eg. Add a new employee

- Delete command
  - Eg. Delete the employee with SSN “99887777”

- Update command
  - Eg. Move the employee “John Smith” to the department 1

INSERT command

- Is used to add 1 or more tuple(s) to a relation

- In order to add a tuple
  - Relation name
  - List of column names
  - List of values for a tuple

INSERT command

- Syntax (one tuple)

INSERT INTO <Table_name>(<List_of_columns>)

VALUES (<List_of_values>)

Example

```
INSERT INTO EMPLOYEE(FName, Minit, LName, SSN)
VALUES ('James', 'E', 'Borg', '888665555')
```

```
INSERT INTO EMPLOYEE(FName, Minit, LName, SSN, Address)
VALUES ('James', 'E', 'Borg', '888665555', NULL)
```

```
INSERT INTO EMPLOYEE
VALUES ('James', 'E', 'Borg', '888665555', '11/10/1937', '450 Stone,
Houston, TX', 'Male', '55000', NULL, 1)
```

INSERT command

- Discussion
  - The order of values is the same to the order of columns
  - The NULL value can be used for non-primary key attributes
  - INSERT command will raise errors if the integrity constraint is violated
    - Primary key
    - Reference
    - NOT NULL constraint

DELETE command

- Is used to remove **tuples** from a relation

- Syntax

```
DELETE FROM <Table_name>
[FROM <List_of_tables>]
[WHERE <Conditions>]
```

Example

```
DELETE FROM EMPLOYEE
WHERE LName= 'Borg'
```

```
DELETE FROM EMPLOYEE
WHERE SSN= '345345345'
```

```
DELETE FROM EMPLOYEE
```

DELETE command

- Discussion
  - The number of removed tuples depends on the condition in WHERE clause
  - A missing WHERE clause specifies that all tuples can be deleted
  - DELETE command can cause the violation of reference constraints
    - Do not permit to remove
    - Remove tuples whose value is being referred
      - CASCADE
    - Set the NULL value to reference values

DELETE command

SSN	FName	Minit	LName	BDate	Address	Sex	Salary	SuperSSN	DNo
123456789	John	B	Smith	01/09/1965	731 Fondren	Male	30000	333445555	5
333445555	Franklin	T	Wong	12/08/1955	638 Voss	Male	40000	888665555	5
999887777	Alicia	J	Zelaya	01/19/1968	3321 Castle	Female	25000	987654321	4
987654321	Jennifer	S	Wallace	06/20/1941	291 Berry	Female	43000	888665555	4
666884444	Ramesh	K	Narayan	09/15/1962	975 Fire Oak	Male	38000	333445555	5
453453453	Joyce	A	English	07/31/1972	5631 Rice	Female	25000	333445555	5
987987987	Ahmad	V	Jabbar	03/29/1969	980 Dallas	Male	25000	987654321	4

ESSN	PNo	Hours
333445555	10	10.0
888665555	20	NULL
453453453	2	10.0
453453453	10	10.0
987654321	30	20.0
999887777	10	10.0

DELETE command

DName	DNumber	MgrSSN	MgrStartDate
Research	5	333445555	05/22/1988
Administration	4	987654321	01/01/1995
Headquarters	1	888665555	06/19/1981

SSN	FName	Minit	LName	BDate	Address	Sex	Salary	SuperSSN	DNo
123456789	John	B	Smith	01/09/1965	731 Fondren	Male	30000	333445555	NULL
333445555	Franklin	T	Wong	12/08/1955	638 Voss	Male	40000	888665555	NULL
999887777	Alicia	J	Zelaya	01/19/1968	3321 Castle	Female	25000	987654321	4
987654321	Jennifer	S	Wallace	06/20/1941	291 Berry	Female	43000	888665555	4
666884444	Ramesh	K	Narayan	09/15/1962	975 Fire Oak	Male	38000	333445555	NULL
453453453	Joyce	A	English	07/31/1972	5631 Rice	Female	25000	333445555	NULL
987987987	Ahmad	V	Jabbar	03/29/1969	980 Dallas	Male	25000	987654321	4

UPDATE command

- Is used to change the value of attributes

- Syntax

```
UPDATE <Table_name>
SET <Attribute_name>=<The_new_value>,
    <Attribute_name>=<The_new_value>,
     ...
[FROM <List_of_table>]
[WHERE <Condition>]
```

Example

```
UPDATE  EMPLOYEE
SET     BDate= '08/12/1965'
WHERE   SSN= '333445555'
```

```
UPDATE  EMPLOYEE
SET     Salary=Salary*1.1
```

UPDATE command

- Discussion
  - Tuples that satisfy conditions in WHERE clause will be modified to the new value
  - A missing WHERE clause specifies that all tuples can be modified
  - UPDATE command can cause violations of the reference constraint
    - Do not allow to modify
    - Modify the values of tuples that are being referred
      - CASCADE

Next lecture

- Data retrieval
  - Basic query
    - **On one schema** : projection & selection

- Read
  - Chapter 5 [The complete book, 2E]
  - Chapter 6, 8 [Fundamentals of Database System, 7E]

- Watch
  - Video clip